MILITARY SPECIFICATION

TERMINALS, LUG: SPLICES, CONDUCTOR: CRIMP STYLE, COPPER, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers insulated and uninsulated crimp-style copper terminal lugs and conductor splices for stranded conductors.

1.2 Classification. Terminals and conductor splices covered by this specification shall be of the following types and classes, as specified (see 6.3):

Type I - Uninsulated
Type II - Insulated
Class 1 - Terminals and conductor splices which conform to all of the requirements of this specification when installed with the crimping tool or crimping dies shown on the applicable MS standard or specification sheet.
Class 2 - Terminals and conductor splices which conform to the material and marking requirements of this specification are replaceable by class 1 terminals, and conform to the performance requirements of this specification when crimped with a tool having crimping dies and motion conforming to the terminal manufacturer's control drawing required by 3.6 (see 6.1.5).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Film (Polyethylene Thin Gage).
QQ-C-502	-	Copper Rods and Shapes; and Flat Products with Finished Edges (Flat Wire, Strips and Bars).
QQ-C-576	-	Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed or Machined Edges, (Plate, Bar, Sheet & Strip).
QQ-S-781	-	Strapping, Steel, Flat and Seals.
TT-I-735	-	Isopropyl Alcohol.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Gleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape, Pressure-Sensitive Adhesive, Waterproof for Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department, Code 93, Naval Air Engineering Center, Lakehurst N.A.S., N.J. 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-T-10727 - Tin Plating, Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engines, Synthetic Base.
MIL-B-43014 - Boxes, Water-Resistant Paperboard, Folding, Set-up and Metal Stayed.
MIL-C-45662 - Calibration System Requirements.

(See Supplement 1 for list of associated specification sheets and MS standards.)

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B75-68 - Seamless Copper Tube.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Detail requirements. Detail requirements or exceptions applicable to particular styles of terminal lugs and conductor splices shall be as specified on the applicable MS standard or specification sheet. In the event of any conflict between requirements of this specification and the requirements of the MS standard or specification sheet, the latter shall take precedence.

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3.2 Qualification. The lug terminals and conductor splices furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.4).

3.3 Material. The material shall be as specified herein. However, when a material not specified is used, it shall be such that the lug terminals and conductor splices conform to the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Metals. The lug terminals and conductor splices shall be fabricated from copper conforming to QQ-C-502, QQ-C-576, ASTM B75-68 or as specified on the applicable MS standard or specification sheet.

3.3.2 Insulation material. The insulation material shall be permanently colored thermoplastic or as specified on the applicable MS standard or specification sheet. Conductor splices need not be totally colored, but a permanent identifying color shall be clearly distinguishable on some portion of the splice. The insulation material shall be noncorrosive, resistant to abrasion and fungus, and shall not support combustion.

3.3.3 Fungus-resistant material. Only fungus-resistant material shall be used on type II terminals. The manufacturer shall submit certification to this effect when submitting type II, class 1 terminals for qualification approval. For type II, class 2 terminals, the certification shall be included in the manufacturer's certified test report to the activity responsible for qualification (see 4.5.2.1).

3.4 Design and construction. Lug terminals and conductor splices shall conform in all respects to the design, dimensions and construction specified herein and on the applicable MS standard or specification sheet. Each terminal size shall be designed for attachment to wire sizes specified on the applicable MS standard or specification sheet by having the terminal receiving barrel reshaped around the conductor. For class 1 terminals and conductor splices, it shall be possible to perform the reshaping operation by means of a crimping tool or crimping dies conforming to the applicable MS standard or specification sheet. For class 2 terminals and conductor splices, it shall be possible to perform this reshaping operation by means of tooling detailed on the manufacturer's control drawings. Terminals and conductor splices shall exhibit no evidence of fracturing or spalling as a result of the reshaping operation. Wire insertion shall be facilitated by bell mouth or chamfer of wire barrel. There shall be no protruding sharp edges of the terminal or conductor splice when reshaping the crimp barrel to the contour of the wire during or after the crimping operation.

3.4.1 Wire facility. Each size lug terminal and conductor splice shall be designed for attachment to the wire diameter range or size specified in the applicable MS standard or specification sheet.

3.4.2 Mounting holes. The tongue of each size lug terminal shall be designed for attachment to the stud size specified. The diameter of the clearance hole for each stud size is listed in table I or on the applicable MS standard or specification sheet.

3.4.3 Wire insulation support. When specified on the applicable MS standard or specification sheet, wire insulation supports shall be required which encompasses the wire insulation and which when deformed by the crimping tool is not required to grip the wire insulation.

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TABLE I. Clearance hole (reference).

Stud size 1/	Clearance hole diameter	
	Minimum	Maximum
2 (.086)	0.090	0.098
4 (.112)	.114	.122
5 (.125)	.129	.137
6 (.138)	.142	.152
8 (.164)	.168	.178
10 (.190)	.193	.203
12 (.216)	.220	.236
1/4 (.250)	.260	.275
5/16 (.312)	.323	.338
3/8 (.375)	.385	.400
7/16 (.437)	.448	.463
1/2 (.500)	.510	.525
5/8 (.625)	.651	.666
3/4 (.750)	.770	.785
7/8 (.875)	.895	.910

1/ Stud size in accordance with Handbook H28.

3.4.4 Insulation. Insulated (type II) lug terminals and conductor splices shall be insulated over the entire outer surface of the barrel and wire insulation support. The insulation, and metal sleeve if used, shall remain in its original position on the barrel of the terminal or conductor splice before and after crimping except as specified in 3.5.5. The insulation shall exhibit no evidence of fracturing or spalling as a result of the reshaping operation.

3.4.5 Finish. Unless otherwise specified (see 3.1), lug terminals and conductor splices shall have conducting parts tin plated over their entire surface in accordance with MIL-T-10727. Bare copper exposed at slug-out points on lug terminals, in strip form, shall not in itself be cause for rejection. The slug-out points shall not exceed 1/8 inch in width. Mercury shall not be used in the manufacture of these terminals.

3.5 Performance requirements. The lug terminals and conductor splices shall conform to the following requirements, when crimped to each of the specified wire sizes with the applicable tool or crimping dies specified in 3.4.

3.5.1 Voltage drop. When tested as specified in 4.7.2, the millivolt drop of the lug terminal or conductor splice shall not exceed the millivolt drop of an equivalent length of wire by more than the value specified in table II.

3.5.2 Current cycling. When tested as specified in 4.7.3, the voltage drop (see 3.5.1) shall be as specified in the applicable columns of table II under "after test".

3.5.3 Dielectric withstanding voltage (type II). When tested as specified in 4.7.4, insulated lug terminals and conductor splices shall show no evidence of damage, arcing, or breakdown.

3.5.4 Salt spray (corrosion). When tested as specified in 4.7.5, lug terminals and conductor splices shall show no evidence of exposure of the base metal or blistering of the plated surfaces. After salt spray, the voltage drop and tensile strength shall be as specified in table II or table III as applicable. For voltage drop, the "after test" requirements shall apply.

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3.5.5 Axial load.

3.5.5.1 Axial load (type I, multiple piece construction, crimped and uncrimped specimens). On multiple piece construction, the metal sleeve on uncrimped lug terminals shall withstand a minimum axial force of 8 pounds and shall not move more than 1/32 inch on the barrel of the lug terminal. The metal sleeve on a crimped lug terminal shall withstand a minimum axial force of 8 pounds (see 4.7.6.1 and 4.7.6.2).

3.5.5.2 Axial load (type II, crimped and uncrimped specimens). The insulation, and metal sleeve if used, on uncrimped lug terminals or conductor splices shall withstand a minimum axial force of 8 pounds and shall not move more than 1/32 inch on the barrel of the lug terminal or conductor splice. The insulation, and metal sleeve if used, on a crimped lug terminal or conductor splice shall withstand a minimum axial force of 8 pounds and shall meet the dielectric withstanding voltage requirement specified in 3.5.3 (see 4.7.6.3 and 4.7.6.4).

3.5.6 Vibration. When tested as specified in 4.7.7.1 or 4.7.7.2, there shall be no evidence of cracking, breaking, or loosening of parts. After vibration, the voltage drop shall not exceed the "after test" values specified in table II, and tensile strength shall be not less than the values specified in table II or table III, as applicable.

3.5.7 Tensile strength. When tested as specified in 4.7.8, lug terminals and conductor splices shall not break or separate from the wire or cable to which it is terminated before the minimum tensile strength specified in table II or table III, as applicable, is reached.

TABLE II. Performance requirements.

Wire size (nominal)	Test current (amperes)	Maximum voltage drop (mV) - Millivolt drop of equivalent length of wire plus				Tensile strength pounds (min.)
		Initial		After test		
		Lug	Splice	Lug	Splice	
26	3	3	6	5	10	7
24	4.5	2	4	4	8	10
22	9	1	2	3	6	15
20	11	1	2	3	6	19
18	16	1	2	3	6	38
16	22	1	2	3	6	50
14	32	1	2	3	6	70
14 1/	32	---	6 1/	---	8 1/	70
12	41	1	2	3	6	110
10	55	1	2	3	6	150
8	73	1	2	3	6	225
6	101	1	2	3	6	300
4	135	1	2	3	6	400
2	181	1	2	3	6	550
1	211	1	2	3	6	650
0	245	2	4	4	8	700
00	283	2	4	4	8	750
000	328	2	4	4	8	825
0000	380	2	4	4	8	875

1/ For MS27429-2, wire size 14, disconnect splice only. All other dash numbers of MS27429 must comply with the requirements of table II.

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3.5.7.1 Additional tensile strength requirements. Table III tensile strength requirements shall apply to the following MS standards:

MS17143	MS21007	MS21012
MS21003	MS21008	MS21013
MS21004	MS21009	MS21014
MS21005	MS21010	MS21015
MS21006	MS21011	

TABLE III. Tensile strength requirements.

Wire size	Navy cable size	Tensile strength pounds (min.)
22-18	1 (1)	15
	1 (7)	16
	1 (10)	15
	1-1/2 (1)	24
	1-1/2 (7)	
	1-1/2 (16)	23
	1-1/2 (41)	24
	2 (7)	
16-14	2-1/2 (1)	40
	2-1/2 (19)	38
	2-1/2 (26)	41
	3 (7)	43
	4 (1)	50
	4 (19)	
	4 (7)	
	4 (41)	
12-10	6 (7)	60
	6 (19)	
	9 (7)	100
	9 (37)	

3.5.8 Immersion (type II). After testing as specified in 4.7.9.1 or 4.7.9.2, lug terminals and conductor splices shall withstand the dielectric withstanding voltage test (see 4.7.4).

3.5.9 Heat aging (type II). After testing as specified in 4.7.10.1 or 4.7.10.2, lug terminals and conductor splices shall meet the dielectric withstanding voltage requirements specified in 3.5.3. Discoloration of the insulation material during this test shall not be cause for rejection.

3.5.10 Flammability (type II). When tested as specified in 4.7.11, lug terminal and splice insulation shall be self-extinguishing within 30 seconds after removal from flame.

3.5.11 Low-temperature crimp (type II). When tested as specified in 4.7.12, insulated lug terminals and splices shall show no evidence of rupture or cracking and shall withstand the dielectric withstanding voltage test (see 4.7.4).

3.5.12 Engaging and disengaging forces and endurance. In addition to those tests specified for conductor splices, MS27429 disconnect splices shall meet the engaging and disengaging values of table IV (see 4.7.13) and the endurance requirements of 4.7.14.

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TABLE IV. Engaging and disengaging forces.

MS27429	Wire size	Engaging force (pounds)		Disengaging force (pounds)	
		Max.	Min.	Max.	Min.
-1	22-18	15	2	15	6
-2	16-14	15	2	15	6
-3	12-10	20	2	20	8

3.6 Manufacturer's control drawing. The terminal manufacturer shall prepare a control drawing for the crimping dies used to crimp his class 2 terminals and conductor splices for the applicable tests. The control drawing shall specify the critical dimensions and motion of the dies and shall also specify the acceptance and in-service gaging requirements. The terminal manufacturer shall certify to the equipment manufacturer that the equipment manufacturer's dies are in accordance with the control drawing and shall furnish to the equipment manufacturer the acceptance and in-service gaging requirements (see 4.5.2.1).

3.7 Identification of product. Class 1 and class 2 terminals and conductor splices for wire sizes 22 and larger shall have the wire size, or range as shown on the applicable MS standard or specification sheet legibly molded or stamped on it in addition to the manufacturer's trademark; additionally, type II terminals and conductor splices shall be color coded in accordance with the applicable MS standard or specification sheet for identification purposes.

3.8 Workmanship. Lug terminals and conductor splices shall be free from burrs, sharp edges, blistering, pitting or peeling of plating, cracks, and other defects which may affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (4.3).
b. Qualification inspection (4.5).
c. Quality conformance inspection (4.6).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table V used in fabricating the lug terminals and conductor splices, are in accordance with the applicable specification or requirements prior to such fabrication.

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TABLE V. Materials inspection.

Material	Requirement paragraph	Applicable specification or requirement
Metals	3.3.1	QQ-C-502, QQ-C-576, or ASTM B75-68
Fungus resistant	3.3.3	MIL-STD-454

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4.1 Assembly to conductors. All lug terminals and conductor splices shall be crimped on the wire by the testing activity, using the specified crimping tool and dies (see 3.1 and 4.5.2.1). The type of wire specified on the applicable MS standard or specification sheet shall be used when performing the specified tests. Where a lug terminal or conductor splice is designed to cover a range of wire sizes, the specified number of sample units for testing shall be selected for both the minimum and maximum wire size within the wire range.

4.4.2 Attachment. During temperature and voltage tests, lug terminals shall be securely bolted back-to-back with a minimum clearance in free air of 18 inches in all directions. Steel bolts, nuts, and washers of the appropriate stud size shall be used. When assembled, the bolt shall not extend more than 1/8 inch beyond the nut (see figure 1).

4.4.3 Temperature stabilization. Voltage drop measurements shall be made after the temperature of the wire has stabilized. Temperature stabilization shall be determined by three consecutive readings within ±1°C at intervals of 3 minutes each.

4.5 Qualification inspection. Qualification inspection shall be performed on sample units produced with equipment and procedures normally used in production at a laboratory acceptable to the Government (see 6.4) for class 1 terminals and conductor splices (see 4.5.1) and in-plant by the contractor for class 2 units (see 4.5.2).

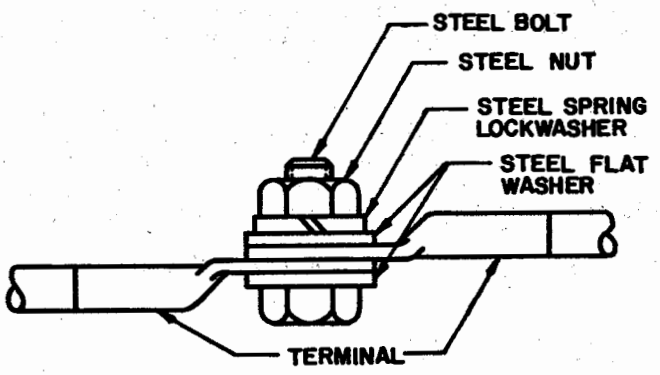


FIGURE 1. Terminal assembly.

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4.5.1 Class 1. Samples for each part number shall be separately packaged and forwarded to the activity responsible for qualification as designated in the letter of authorization (see 6.4). Each sample shall be identified by marking each package with the following information:

Sample for qualification
Specification MIL-T-7928G
MS or specification sheet part number _____ (class 1 items)
TERMINALS, LUG; OR SPLICE, CONDUCTOR: (as appropriate)
CRIMP STYLE, COPPER
Type _____ Class _____
Name of contractor
Contractor's part number
Submitted (date) under authiorization (reference letter authorizing the inspection)

4.5.2 Class 2. Qualification inspection shall be performed by the contractor and the test report shall be furnished to the activity responsible for qualification (see 6.4). Upon approval by the activity responsible for qualification, the control drawing number for the crimping dies (see 3.6) shall be listed in the applicable qualified products list.

4.5.2.1 Test reports. The contractor shall furnish the activity responsible for qualification with a certified test report, in duplicate, showing quantitative results for class 2 terminals and splices as required by this specification and further certified in the forwarding letter by an officer of the contractor. The report shall designate the equivalent part number of the class 1 terminal or conductor splice which will replace the class 2 terminal or conductor splice submitted. The report shall also include the manufacturer's control drawings specified in 3.6 and the part numbers of the tool and die used to perform the qualification tests.

4.5.3 Sample size. The number of lug terminals or conductor splices specified in table VI for each barrel size of each military standard (see 3.1) for which qualification is sought shall be submitted for qualification inspection. In addition, five lug terminals for each stud size for which qualification is sought shall be submitted for visual and dimensional examination. Fifteen additional class 1 lug terminals or conductor splices shall be submitted to the activity responsible for qualification for any additional testing deemed necessary. The sample size shall be the same for class 1 and class 2 terminals and conductor splices.

4.5.4 Inspection routine. Sample units shall be subjected to the qualification inspection specified in table VI, in the order shown. All sample units shall be subjected to the inspection of group I. The sample units shall then be divided into groups II through XI, as shown in table VI, and subjected to the inspection for their particular group.

4.5.5 Failures. One or more failures shall be cause for refusal to grant qualification approval.

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TABLE VI. Qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Sample units Type I	Sample units Type II
Group I				
Visual and dimensional examination	3.1, 3.3, 3.4, 3.7, and 3.8	4.7.1	22 12 3/	40 55 1/ 52 2/
Group II				
Voltage drop	3.5.1	4.7.2	4	4
Current cycling	3.5.2	4.7.3		
Voltage drop	3.5.1	4.7.2		
Group III				
Voltage drop	3.5.1	4.7.2	4	4
Vibration	3.5.6	4.7.7		
Voltage drop	3.5.1	4.7.2		
Tensile strength	3.5.7	4.7.8		
Group IV				
Immersion (105°C)	3.5.8	4.7.9.1	---	8
Immersion (150°C)	3.5.8	4.7.9.2		20 2/
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		
Group V				
Flammability	3.5.10	4.7.11	---	2
Group VI				
Voltage drop	3.5.1	4.7.2	4	4
Salt spray (corrosion)	3.5.4	4.7.5		
Voltage drop	3.5.1	4.7.2		
Tensile strength	3.5.7	4.7.8		
Group VII				
Heat aging (105°C)	3.5.9	4.7.10.1	---	4
Heat aging (150°C)	3.5.9	4.7.10.2	---	4 2/
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		
Group VIII				
Low-temperature crimp (type II)	3.5.11	4.7.12	---	4
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		
Group IX				
Axial load (type I)	3.5.5.1	4.7.6.1 and 4.7.6.2	10 4/	
Axial load (type II)	3.5.5.2	4.7.6.3 and 4.7.6.4		10
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		
Group X 1/				
Engaging and disengaging forces	3.5.12	4.7.13	---	10

See footnotes at end of table.

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TABLE VI. Qualification inspection - Continued.

Inspection	Requirement paragraph	Method paragraph	Sample units Type I	Sample units Type II
Group XI 1/				
Voltage drop	3.5.1	4.7.2	---	5
Endurance	3.5.12	4.7.14	---	
Voltage drop	3.5.1	4.7.2		

1/ Applicable to MS27429 splices only.
2/ Applicable to 150°C items only (see 3.1).
3/ Applicable to one-piece construction only.
4/ Applicable to multiple-piece construction only.

4.5.6 Retention of qualification. To retain qualification, the contractor shall forward a report at 12 month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery, group A, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
b. A summary of the results of tests performed for periodic inspection, groups B and C, including the number and mode of failures. The summary shall include results of all group A quality conformance inspection performed and completed during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicated failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative item of those covered by a single specification sheet to testing in accordance with the qualification inspection requirements.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.2 Inspection lot. An inspection lot, as far as practicable, shall consist of all lug terminals and conductor splices of a single type, class, size, and composition manufactured under essentially the same conditions and offered for inspection at one time.

4.6.3 Group A inspection. Group A inspection shall consist of the examination and tests specified in table VII and shall be made on the same set of sample units, in the order shown. After the visual examination is completed, the lot shall be divided and tensile strength tests performed on both the minimum and the maximum diameter wires within the accommodated wire range.

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4.6.3.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for special inspection levels. The acceptable quality level (AQL) shall be as specified in table VII. Major and minor defects shall be as defined in MIL-STD-105 and table X.

4.6.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE VII. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	AQL (percent defective)		Sampling plan
			Major	Minor	
Visual and mechanical examination	3.1, 3.3, 3.4, 3.7, and 3.8	4.7.1	1.0	4.0	S-4
Tensile strength	3.5.7	4.7.8	0.65	---	S-1

4.6.4 Periodic inspection. Periodic inspection shall consist of groups B and C. Except when the results of these inspections show noncompliance with the applicable requirements (see 6.5), delivery of products which have passed group A shall not be delayed pending the results of these periodic inspections.

4.6.4.1 Group B inspection. Group B inspection shall consist of the tests specified in table VIII, in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

TABLE VIII. Group B inspection.

Inspection	Requirement paragraph	Method paragraph
Group 1 (15 units) (type II only)		
Dielectric withstanding voltage	3.5.3	4.7.4
Engaging and disengaging forces 1/	3.5.12	4.7.13
Group 2 (15 units) (any type)		
Voltage drop	3.5.1	4.7.2

1/ Applicable to MS27429 splices only.

4.6.4.1.1 Sampling plan. Fifteen type I and 30 type II lug terminals or conductor splices, as applicable, shall be selected from those covered by a single MS standard or specification sheet and having the same barrel size, 12 months after the date of notification of qualification, and after each subsequent 12 month period. A manufacturer's normal quality control tests and production tests may be used to fulfill all or part of group B inspection; however, all of the group B inspection shall be completed as specified.

4.6.4.1.2 Failures. If one or more units fail to pass group B inspection, the sample shall be considered to have failed.

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4.6.4.1.3 Disposition of samples. Samples which have been subjected to group B inspection shall not be delivered on the contract or order.

4.6.4.2 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table IX, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

4.6.4.2.1 Sampling plan. As specified in 4.5.3, the number of sampling units of each wire size and type shall be selected from production lots 36 months after the date of notification of qualification, and after each subsequent 36-month period. In those cases where a terminal or conductor splice covers a range of wire sizes, the number of sample units shall be selected for a maximum and an equal number for the minimum wire size within the range.

4.6.4.2.2 Failures. If one or more units fail to pass group C inspection, the sample shall be considered to have failed.

4.6.4.2.3 Disposition of sample units. Sample units which have been subjected to group C inspection, shall not be delivered on the contract or order.

TABLE IX. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Sample units Type I	Sample units Type II
Group I				
Current cycling	3.5.2	4.7.3	4	4
Voltage drop	3.5.1	4.7.2		
Group II				
Vibration	3.5.6	4.7.7	4	4
Voltage drop	3.5.1	4.7.2		
Tensile strength	3.5.7	4.7.8		
Group III				
Immersion (105°C)	3.5.8	4.7.9.1		8
Immersion (150°C) 2/	3.5.8	4.7.9.2		20
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		
Group IV				
Flammability	3.5.10	4.7.11		2
Group V				
Salt spray (corrosion)	3.5.4	4.7.5	4	4
Voltage drop	3.5.1	4.7.2		
Tensile strength	3.5.7	4.7.8		
Group VI				
Heat aging (105°C)	3.5.9	4.7.10.1		4
Heat aging (150°C) 2/	3.5.9	4.7.10.2		4
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		

See footnotes at end of table.

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TABLE IX. Group C inspection - Continued.

Inspection	Requirement paragraph	Method paragraph	Sample units Type I	Sample units Type II
Group VII				
Low-temperature crimp (type II)	3.5.11	4.7.12		4
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		
Group VIII				
Axial load (type I)	3.5.5.1	4.7.6.1 and 4.7.6.2	10 3/	
Axial load (type II)	3.5.5.2	4.7.6.3 and 4.7.6.4		10
Dielectric withstanding voltage (type II)	3.5.3	4.7.4		
Group IX				
Engaging and disengaging forces 1/	3.5.12	4.7.13		10
Group X				
Endurance 1/	3.5.12	4.7.14		5
Voltage drop	3.5.1	4.7.2		

1/ Applicable to MS27429 splices only.
2/ Applicable to 150°C items only (see 3.1).
3/ Applicable to multiple piece construction only.

4.6.5 Inspection of preparation for delivery. Except when commercial packaging is specified, the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract or purchase order (see 6.3).

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. Lug terminals and conductor splices shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with this specification and applicable military standard or specification sheet. Dimensions shall conform to those specified on the manufacturer's drawing for class 2, and to the applicable MS standard or specification sheet for class 1. Classification of defects for visual and mechanical examination shall be as specified in table X.

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TABLE X. Classification of defects.

Examination of product	Major	Minor
Dimensions:		
Overall length (max.)	X	---
Width of tongue (max.)	X	---
Stud hole (min.)	X	---
Centerline of stud hole to end of tongue (max.)	X	---
Washer clearance	X	---
All other dimensions	---	X
Materials	X	---
Workmanship and marking	---	X

4.7.2 Voltage drop (see 3.5.1). Lug terminals and conductor splices shall be tested as follows:

a. Test points: Lug terminal measurements shall be made by puncturing the insulation of the current-carrying conductor 1/16 inch back from the wire-receiving end of the terminal for one test point and using the intersection of the tongue and barrel for the other test point. The distance between the two test points shall be noted.

Conductor splice measurements shall be made by puncturing the insulation of the current-carrying conductor on each end of the splice 1/16 inch back from the wire-receiving end of the splice. The distance between the two test points shall be noted.

Measurement of the current-carrying conductor shall be made by puncturing the conductor insulation the same distance between test points as that noted for the terminal and splice measurements. The conductor measurement shall be made midway on the wire between termination joints. The millivolt drop of the equivalent length of wire may be determined by averaging four readings taken on 10-inch lengths of wire selected at random throughout the supply of wire to be used for subsequent tests.

b. Measurements: The millivolt drop through the crimp termination and the current-carrying conductor shall be measured while the specified test current (see table II) is being applied and after the temperature of the wire has stabilized (see 4.4.3).

4.7.3 Current cycling (see 3.5.2). Test samples attached to 3-foot lengths of appropriate wire shall be subjected to 50 current cycles. Each cycle shall consist of 30 minutes at 125 percent of the test current specified in table II, followed by 15 minutes at no load. Voltage drops shall be measured at test currents specified in table II after the test assembly has returned to room temperature.

4.7.4 Dielectric withstanding voltage (type II) (see 3.5.3). Insulated lug terminals and conductor splices shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Preparation of specimen - The tongue of the type II crimped lug terminal shall be sealed with a suitable insulating compound to a depth suffie cient to close the open end of the lug barrel without covering the depression resulting from the applied crimping pressure. One of the wires in a type II crimped splice shall be cut as close to the insulating sleeve as possible, and the wire entrance end shall be insulated as

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described above. The samples shall be immersed in a 5 percent salt water solution to a depth sufficient to cover the crimped areas of the barrel and insulating grip.

b. Magnitude of voltage - 1,500 volts.

c. Nature of potential - AC.

4.7.5 Salt spray (corrosion) (see 3.5.4). The salt spray (corrosion) test shall be in accordance with method 101, test condition B of MIL-STD-202. Terminal lugs shall be crimped to short lengths of proper size wire and mounted in the salt spray (corrosion) chamber in a horizontal position. Conductor splices shall be crimped to short lengths of wire and free ends of the wire shall be terminated with approved lug terminals. Adjoining samples shall be separated by at least 1/4 inch. During this test, the samples shall not come in contact with metallic or wooden objects, and the salt spray (fog) shall be a salt solution concentration of 5 percent and shall have free access to the samples. The samples shall be removed from the salt spray, washed with distilled water, air dried for a minimum of 1 hour and examined for compliance with 3.5.4. Prior to subsequent testing, the corrosion on the terminal's tongue shall be removed. The terminal lugs and conductor splices shall then be subjected to the voltage drop (see 4.7.2) and tensile strength (see 4.7.8) tests.

4.7.6 Axial load.

4.7.6.1 Axial load (type I, multiple piece construction, uncrimped specimens) (see 3.5.5.1). Five uncrimped specimens shall have a 0.022 inch diameter hole drilled through the metal sleeve and a piano wire of 0.020 inch diameter shall be inserted through the drilled hole. The specimens shall then be fastened in a tensile testing machine and the axial force specified in 3.5.5.1 shall be exerted on the piano wire (see figure 2). An alternate test method may be used provided the requirements of 3.5.5.1 are met.

4.7.6.2 Axial load (type I, multiple piece construction, crimped specimens) (see 3.5.5.1). Five new specimens shall be crimped to the specified smallest size wire accommodated by the terminal lug. The specimens shall be tested by applying the specified axial force (see 3.5.5.1) at a maximum speed of one inch per minute between the crimped conductor and a suitable test jig positioned under the metal sleeve. The free end of the crimped conductor and the test jig shall be fastened in the jaws of the tensile machine (see figure 3).

4.7.6.3 Axial load (type II, uncrimped specimens) (see 3.5.5.2). Five uncrimped specimens, each of type II terminal lugs, and conductor splices shall have four 0.022 inch diameter holes drilled through the insulation overhang, and metal sleeve if used, and four piano wires of 0.020 inch diameter shall be inserted through the drilled holes. The specimens shall then be fastened in a tensile testing machine and the axial force specified in 3.5.5.2 shall be exerted on the piano wires (see figure 2). An alternate test method may be used provided the requirements of 3.5.5.2 are met.

4.7.6.4 Axial load (type II, crimped specimens) (see 3.5.5.2). Five new specimens each of the terminal lugs and conductor splices shall be crimped to the specified smallest size wire accommodated by the terminal lug or splice. The specimens shall then be subjected to humidity (steady state) in accordance with test condition B, method 103, of MIL-STD-202. After completion of the humidity conditioning, prior to axial load, (one conductor of the splice shall be cut as close to the insulation sleeve as possible) the terminal lug and conductor splice specimens shall be tested by applying the specified axial force (see 3.5.5.2) at a maximum speed of one inch per minute between the crimped conductor and a suitable test jig positioned under the insulation overhang. The free end of the crimped conductor and the test jig shall be fastened in the jaws of the tensile machine (see figure 3). Each crimped terminal lug or conductor splice shall then be tested for dielectric withstanding voltage as specified.

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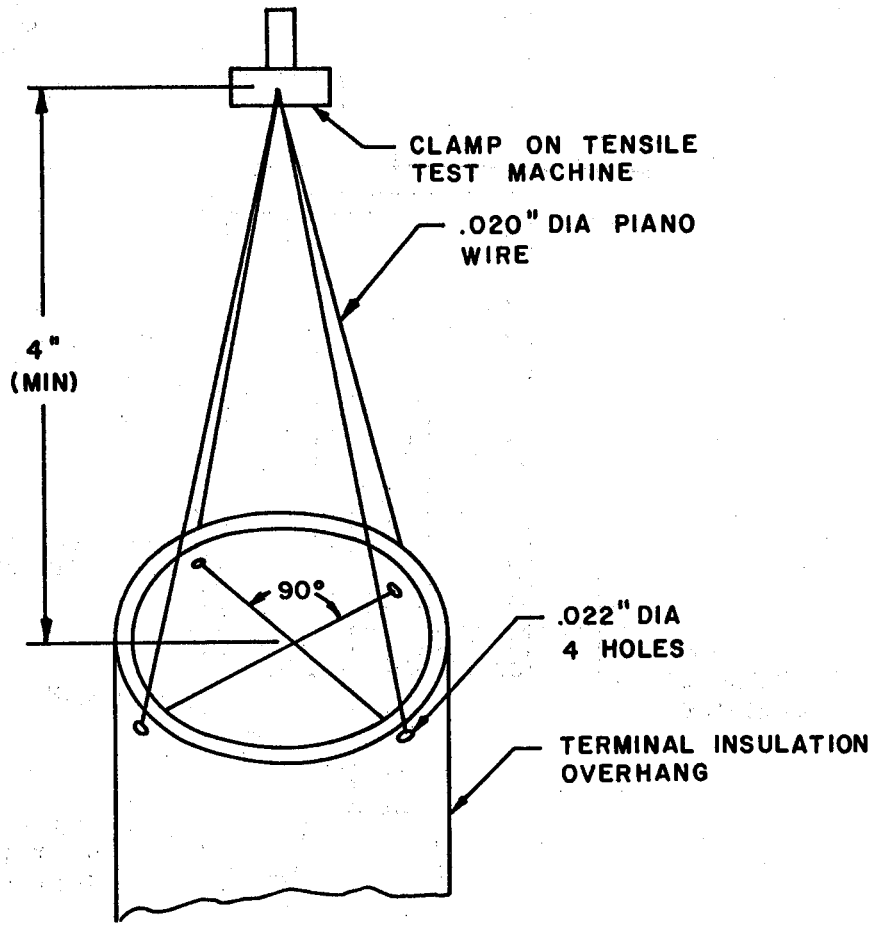


FIGURE 2. Set up for axial load test (uncrimped specimens).

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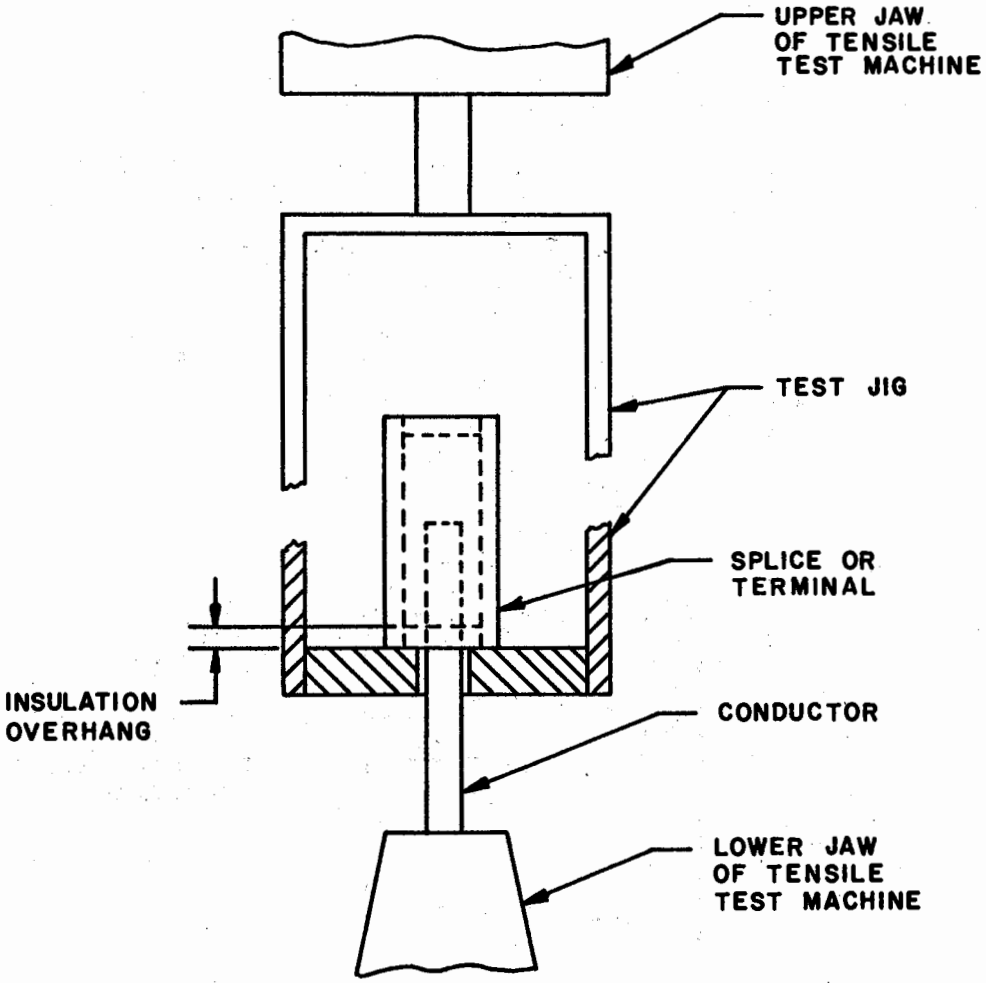


FIGURE 3. Set up for axial load test (crimped specimens).

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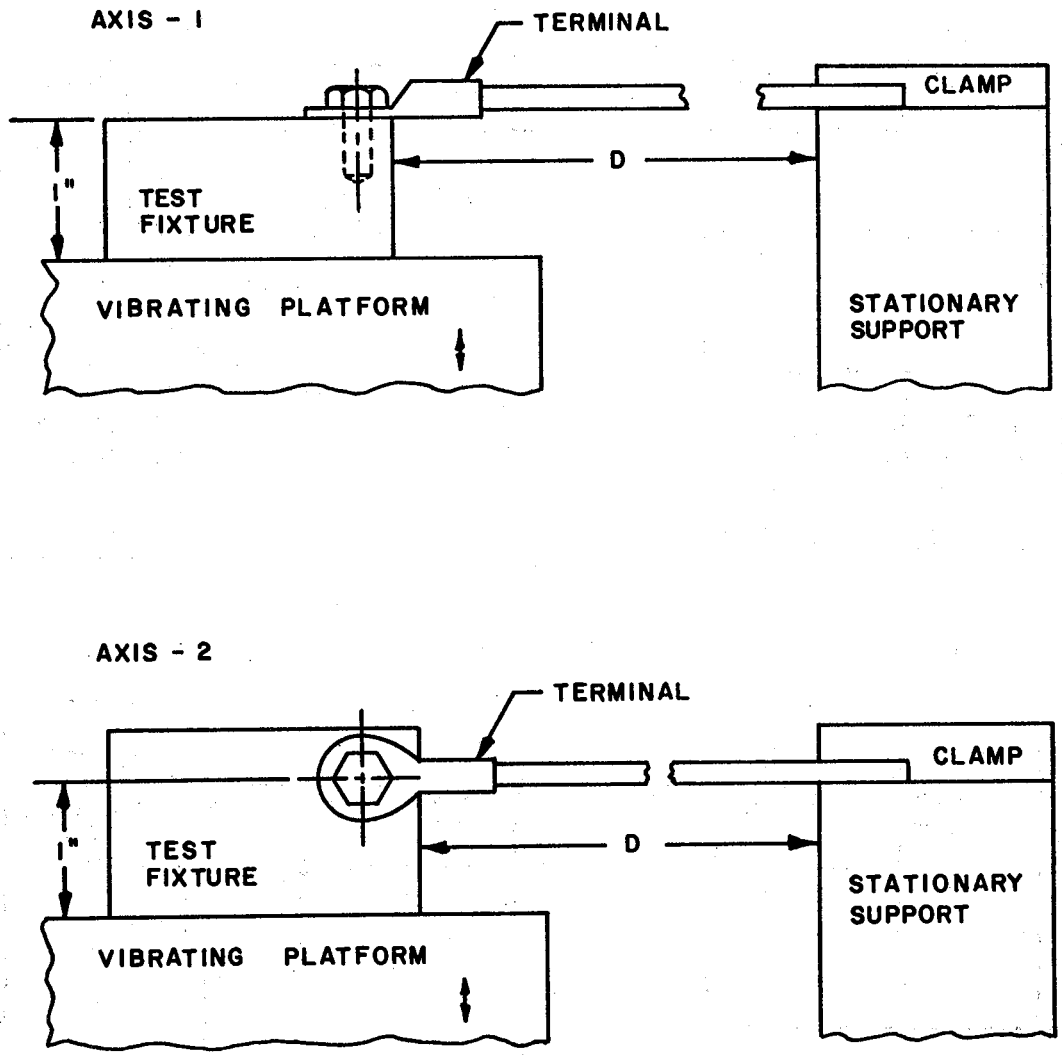


FIGURE 4. Vibration test for lug terminals.

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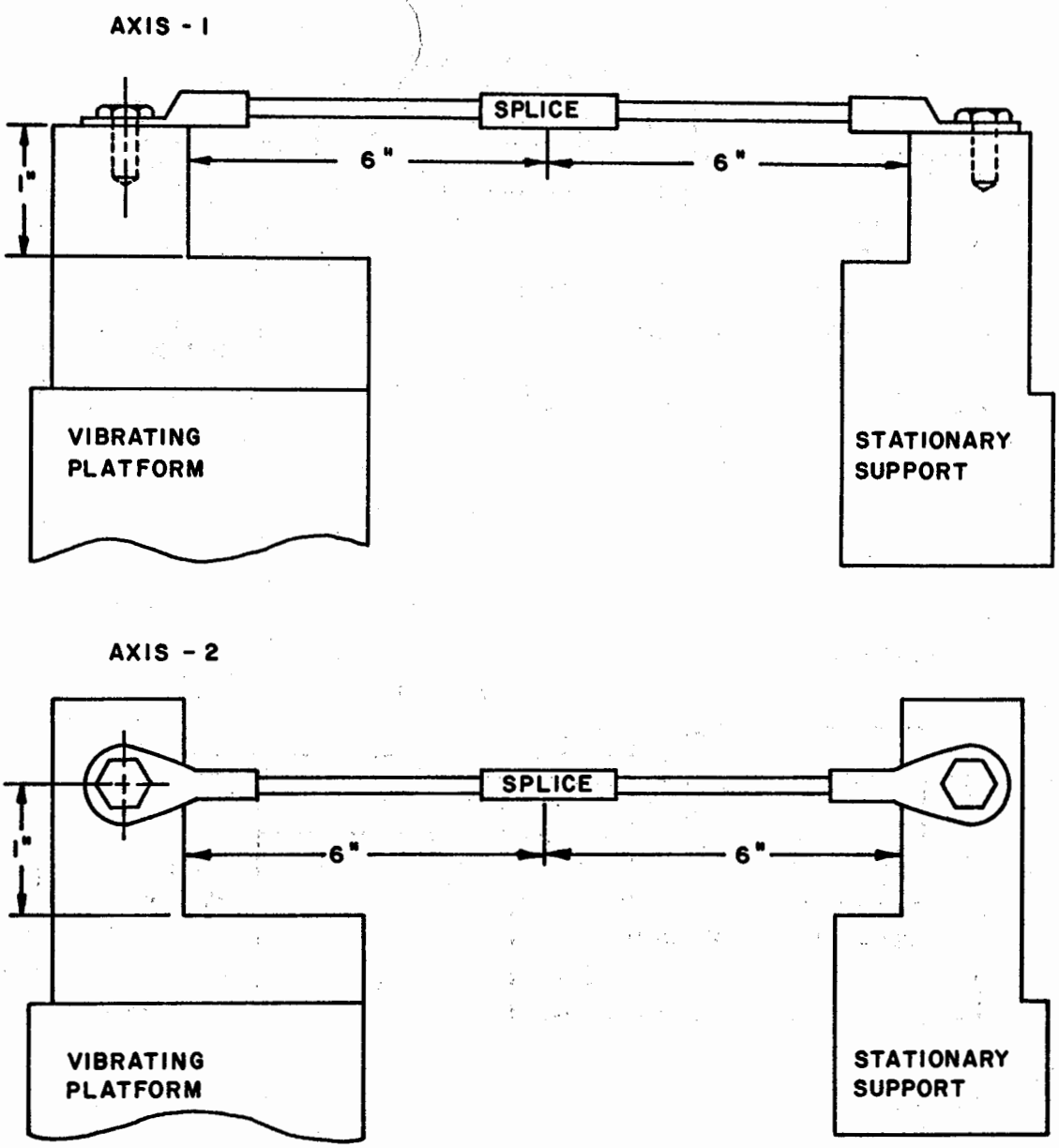


FIGURE 5. Vibration test for conductor splices.

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4.7.7 Vibration.

4.7.7.1 Vibration (with insulation support and sizes 12 and larger without insulation support (see 3.5.6). Lug terminals and conductor splices shall be attached to a length of wire. The terminals shall be rigidly mounted to and not more than 1 inch above the vibrating platform by their normal mounting means (see figure 4). The test terminals shall be so mounted that the junction of the other end of the test sample shall be secured to a stable support 12 inches external to the vibrating table with all slack or tension removed from the wire. Conductor splices shall be assembled and so mounted that the center of the splice is 6 inches from the vibrating platform and 6 inches from the support (see figure 5). Lug terminals shall be attached to the opposite ends of the wires attached to the conductor splice to insure equal distribution of stress among the strands. The specimens shall be vibrated in accordance with method 201 of MIL-STD-202 for 18 hours on each of two axes mutually perpendicular to each other and to the axis of the wire. Following the test, the lug terminal and conductor splice shall be subjected to the voltage drop (see 4.7.2) and tensile strength (see 4.7.8) tests.

4.7.7.2 Vibration (lug terminals sizes 14 and smaller without insulation support (see 3.5.6). Lug terminals shall be attached to a length of wire. The terminals shall be rigidly mounted to a test fixture 1 inch in height, securely fastened to the vibrating platform (see figure 4). The test terminals shall be so mounted that the junction of the terminal tongue and barrel lies directly at the edge of the test fixture. The opposite end of the wire shall be clamped securely to a stationary support external to the vibrating platform and 17 inches (±1/2 inch) from the test fixture with only enough tension applied to remove the slack from the wire. The test fixture shall then be moved toward the stationary support (or vice versa) a distance of 1/8 inch (±1/32 inch) to put a controlled amount of slack in the wire. The terminals shall be vibrated in accordance with method 201 of MIL-STD-202 for 2 hours in each of two axes mutually perpendicular to each other and to the axis of the wire. Following the test, the lug terminals shall be subjected to the voltage drop (see 4.7.2) and tensile strength (see 4.7.8) tests.

4.7.8 Tensile strength (see 3.5.7). Lug terminals and conductor splices attached to suitable wires (see 4.4.1) shall be placed in a standard tensile-testing machine and sufficient force applied to pull the wire out of the sample or break the wire or sample. The travel speed of the head shall be 1 inch per minute. The clamping surfaces may be serrated to provide sufficient clamping force.

4.7.9 Immersion.

4.7.9.1 Immersion (type II) (105°C temperature) (see 3.5.8). Insulated lug terminal and conductor splice samples shall be divided into equal parts, one part immersed in aviation hydraulic fluid conforming to MIL-H-5606, and the other part immersed in aircraft lubricating oil conforming to MIL-L-7808 for a period of 20 hours. Upon removal from the fluids, the terminals and splices shall be air-dried for 1 hour at room temperature, after which time any remaining excess fluid may be wiped from the sample. The samples shall then be subjected to the dielectric withstanding test (see 4.7.4).

4.7.9.2 Immersion (type II) 150°C temperature (see 3.5.8).

a. Insulated lug terminals and conductor splice samples shall be divided into equal parts and immersed in fluids listed below at room temperature for a period of 20 hours. Upon removal from the fluids, samples are air-dried for one hour and then any remaining fluid may be wiped from the samples. The samples then shall be subjected to the dielectric withstanding voltage test (see 4.7.4).
b. Same as a., except fluids for test to be at 120°F.

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c. Fluids for test.
   1. Room temperature - JP-4 or JP-5 } MIL-T-5624
      TT-I-735
   2. 120°F - MIL-H-5606
      SKY DROL - 500B
      MIL-L-7808

4.7.10 Heat aging.

4.7.10.1 Heat aging (type II) 105°C temperature (see 3.5.9). Insulated lug terminals and conductor splices attached to suitable wires shall be conditioned in a heat chamber at a temperature of 121° ±1°C for a period of 120 hours. After removal from the chamber, the specimens shall be cooled to a temperature of 23° ±3°C within a period of 1 hour. When cool, the specimens shall be subjected to the dielectric withstanding voltage test (see 4.7.4).

4.7.10.2 Heat aging (type II) 150°C temperature (see 3.5.9). Insulated lug terminals and conductor splices attached to suitable wires shall be conditioned to a heat chamber at a temperature of 175° ±5°C for a period of 120 hours. After removal from the chamber, the specimens shall be cooled to a temperature of 23° ±3°C within a period of 1 hour. When cooled, the specimens shall be subjected to the dielectric withstanding voltage test (see 4.7.4).

4.7.11 Flammability (type II) (see 3.5.10). Insulated lug terminals and conductor splices attached to suitable conductors shall be suspended in a draft-free enclosure above a Bunsen burner with a wing-top flame spreader. The tip of a 2-inch gas flame with an inner core one-third its height shall be applied for a period of 20 seconds to one-half the length of insulation. The Bunsen burner shall have a 1/4-inch inlet, a nominal bore of 3/8 inch, and a length of approximately 4 inches above the primary inlets. The wing-top flame spreader shall have a 1/16-inch by 2-inch opening.

4.7.12 Low-temperature crimp (type II) (see 3.5.11). Insulated lug terminals and conductor splices, suitable wires (see 4.4.1) and the crimping tool shall be maintained at a temperature of -5° ±1°C for a period of 1 hour. The specimens shall be crimped to the proper size wire while still at -5° ±1°C. The specimens shall then be exposed to -65° ±1°C, and maintained at that temperature for a period of 1 hour. When the specimens have reached room temperature, they shall be subjected to the dielectric withstanding voltage test specified in 4.7.4.

4.7.13 Engaging and disengaging (see 3.5.12). Sample splices not previously used shall be engaged and disengaged ten times.

4.7.14 Endurance (see 3.5.12). The disconnect splices shall be subjected to 100 cycles of mating and unmating at a uniform rate of 8 to 10 cycles per minute. After completion of this test, the splices shall meet the millivolt drop test of table II, as applicable.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, or as specified (see 6.3).

5.1.1 Level A.

5.1.1.1 Cleaning. Terminal lugs and conductor splices shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Terminal lugs and conductor splices shall be dried in accordance with MIL-P-116.

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5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.3), terminal lugs and conductor splices shall be unit packaged ten each in accordance with MIL-P-116, method III insuring compliance with the applicable requirements of that specification.

5.1.1.5 Intermediate packaging. Terminal lugs and conductor splices packaged as specified in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packages, not to exceed 100 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 100 unit packages.

5.1.2 Level C. Terminal lugs and conductor splices shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, or as specified (see 6.3).

5.2.1 Level A. The packaged terminal lugs and conductor splices shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seals, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged terminal lugs and conductor splices shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged terminal lugs and conductor splices shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.5.

5.4.3 Army procurements.

5.4.3.1 Level A (maximum military protection) unit and intermediate packaging. All unit and intermediate containers shall either be weather (or water) resistant or overwrapped with waterproof barrier materials. Containers conforming to PPP-B-566 or PPP-B-676 shall be overwrapped with waterproof barrier materials or shall conform to MIL-B-43014 (see 5.1.1.4 and 5.1.1.5).

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5.4.3.2 Level A (maximum military protection) and level B (minimum military protection) packing. For level A packing the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2 style 4 of PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.3.3 Commercial packaging. Commercial packaging (including unit and intermediate packaging, packing and marking) shall be in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. Terminals and splices covered by this specification are for use on all conductors having diameters within the ranges specified on the applicable MS standard or specification sheet.

6.1.1 Conductor splices. Permanent type or disconnect type conductor splices are primarily intended for use in splicing two conductors.

6.1.2 Type I. Type I terminals and conductor splices are intended for applications when total temperature of the conductor does not exceed 175°C.

6.1.3 Type II. Type II terminals and conductor splices are intended for applications where total temperature of the conductor does not exceed 105°C (150°C for higher temperatures) and where the insulation of the barrel is a major requisite.

6.1.4 Class 1. Class 1 terminals and conductor splices are intended for use by military equipment manufacturers and Service activities. These terminals and conductor splices will be the only style stocked by the Services and supplied to their activities. Tools for crimping class 1 terminals and conductor splices are listed on the applicable MS standard or specification sheet.

6.1.5 Class 2. Class 2 terminals and conductor splices are considered as conforming to this specification only when crimped with a tool having crimping dies and motion conforming to the applicable manufacturer's control drawing listed in QPL-7928. These terminals will not be procured or stocked by DOD or supplied to the Services.

6.1.6 Installation. Where class 2 terminals are used, the installation will be such that class 1 terminals will be adequate as a replacement without rework of the installation or terminal. Parts list will indicate the class 1 terminal that will be used for Service replacement of class 2 terminals if used.

6.1.7 Hand crimping tools. Hand crimping tools conforming to the applicable MS standard or specification sheet and as approved by the activity responsible for qualification will be stocked by the Services for field maintenance. Dies for the crimping tools should be checked at least once a month.

6.1.8 Hand or power tools (with crimping dies). Hand or power tools used by the military equipment manufacturers to crimp class 2 terminals must have crimping dies conforming to the terminal manufacturer's control drawing as listed in QPL-7928. It is the responsibility of the equipment manufacturer to determine that the crimping dies in his tools conform to this control drawing. These tools are not be be used for overhaul and maintenance by the Services.

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6.1.9 Hand or power tools (with forming dies). Hand or power tools having forming dies complying with the forming dimensions of the tools qualified by the Services as acceptable for class 2 terminals, are acceptable for military equipment manufacturers without additional qualification. These tools are not acceptable for overhaul or maintenance purposes by the Services.

6.2 Drawing requirements. The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with MIL-D-1000.

6.3 Ordering data. Procurement documents should specify:

(a) Title, number and date of this specification.
(b) Type and class of lug terminal or splice conductor desired and applicable MS or specification sheet part number (MS or specification sheet part number will not be applied to class 2 terminals (see 1.2).
(c) Inspection of commercial packaging (see 4.6.4).
(d) Levels of preservation-packaging and packing required (see 5.1 and 5.2). Quantity per unit package, if other than ten each (see 5.1.1.4).
(e) Special marking, if required (see 5.3).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Air Systems Command, Navy Department, Washington, DC 20361. However, the Commanding Officer, Naval Weapons Support Center, Code 3074, Crane, Indiana 47522, has been designated agent for the Naval Air Systems Command to establish the QPL. Information pertaining to qualification of products may be obtained from the Naval Air Systems Command Agent.

6.4.1 Class 2 terminals. When requested by the activity responsible for qualification, manufacturers desiring approval of class 2 terminals should make available to the designated testing activity the tooling necessary for performing the crimping operation. At the conclusion of the testing, the manufacturer's tooling will be returned.

6.5 Noncompliance. If a sample fails to pass group B or C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B or C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstituted in the event of failure of group B inspection; however, final acceptance and shipment shall be withheld until the group B or C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreements (ASCC AIR STDS 12/4, 12/13, and STANAG 3314). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.7 Superseding data. This specification supersedes MIL-T-7928F, dated 2 October 1967, NUK-E-16366D(SHIPS) (in part), dated 23 June 1959.

Custodians:
Army - EL
Navy - AS
Air Force - 85

Review activities:
Army - AV, EL, MI, MU, SG, AT
Navy - SH
Air Force - 80
DSA - GS

User activities:
Army - AT
Navy - OS
Air Force - 17

International interest (see 6.6)

Preparing activity:
Navy - AS

Agent:
DSA - ES

(Project 5940-0767)

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